

National College of Ireland

Project Submission Sheet

Student Name: Varun Sai Yandapalli
Student ID: 23325836
Programme: MSCDAD_B **Year:** 2nd sem
 Business Intelligence and Business Analytics
Module: Pritesh Tiwari
Lecturer:
Submission Due Date: 20th April 2025
Project Title: BIBA CA_2

Word Count: 6740

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Signature: Y Varun Sai
Date: 20th April 2025

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

Office Use Only	
Signature:	
Date:	

Penalty Applied (if applicable):	
----------------------------------	--

AI Acknowledgement Supplement

Data Mining and Machine Learning - 1

Data Mining and Machine Learning Project

Your Name/Student Number	Course	Date
Varun Sai Yandapalli 23325836	- MSCDAD_B	20 th April 2025

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

SPECIFICATION REPORT

Section 1: Background Information, Business Goals & Exploratory Data Analysis

Company Background

Drivers of global market leadership belong to Adidas operating in the athletic footwear apparel and accessory sector. The company reaches its customers through retail platforms consisting of physical stores with outlets as well as a fully developed online e-commerce system. International market penetration for Adidas happens through gaining brand strength using advanced product production capabilities. United States market areas reveal wide-ranging discrepancies between business performance data across regional business units at present.



Business Problem

Data shows that the U.S. sales performance of Adidas remains below average in both Midwest and southeastern states while most successful markets stem from West and Northeast regions. Such weaker markets experience multiple operational issues that result in performance problems.

- Lower overall sales volumes
- Limited penetration of online sales channels
- Volatile or low operating margins

Target market regions received poor customer loyalty evaluation results because Adidas implemented ineffective customer engagement programs within those areas. The markets need urgent attention for improved customer targeting approaches combined with strong engagement methods alongside sales management systems according to data analysis.

Business Goals

Adidas set a focused set of business goals as a response to these challenges drive operational and strategic improvement.

- Increase sales performance and profitability in underperforming U.S. regions
- Targeted CRM initiatives should be used to strengthen customer engagement and loyalty.
- Use data driven dashboards and automatically CRM workflows to enable personalised outreach and track the effectiveness of the campaign in real time.

My Role and Project Approach

At Business Intelligence and Analytics, I was a key player between data infrastructure and customer relationship systems. I was involved in several functional area of the project including (but not limited to):

- The regional sales pattern analysis was conducted using Power BI.
- We have simulated customer loyalty and engagement behavior using the generated data using the platform Mockaroo.
- Drew up and executed automated CRM campaign logic in Salesforce through the use of custom fields, validation rules, and Flow Builder
- Structured and uploaded to relational datasets to PostgreSQL and support scalable analytics integration.
- Built and executed interactive dashboards to illustrate sales trends, regional CRM outcomes and loyalty metrics

By adopting an end-to-end strategy, data unification was easy, marketing processes were automated and strategic insights were used to guide Adidas' regional sales initiative.

Exploratory Data Analysis (EDA)

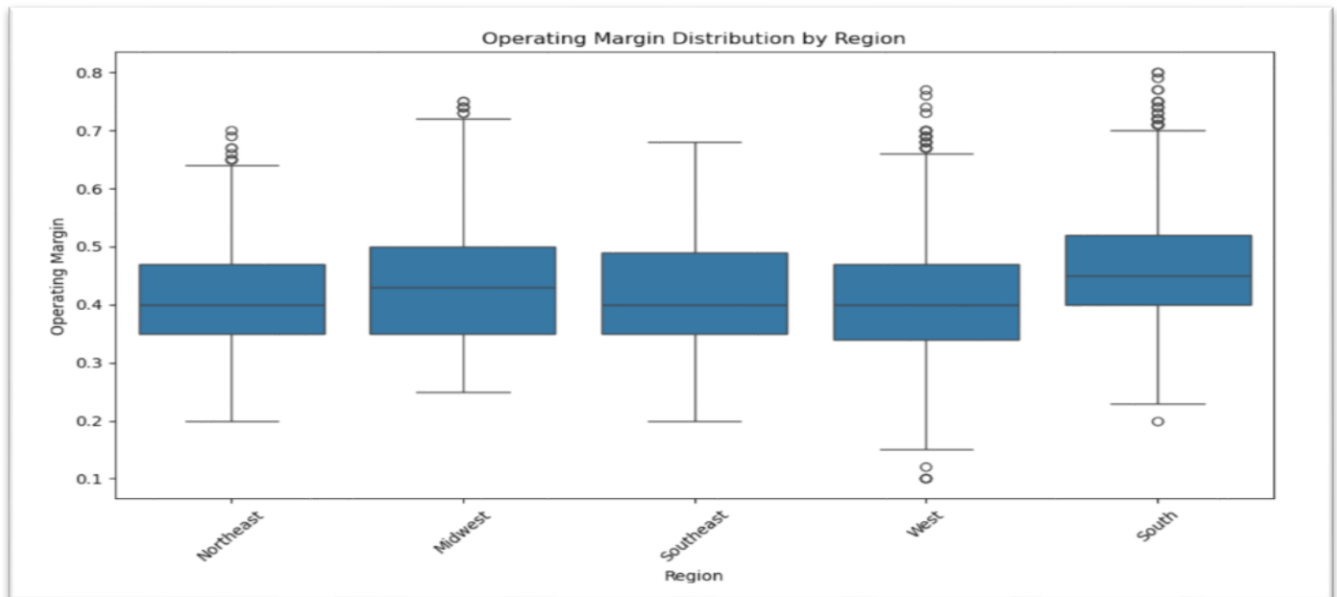
Datasets were imported and analyzed using Python with extracted ones on PostgreSQL. The following was the insights from the EDA:

1. Missing Values & Data Types

Hence, it was ensured that the dataset for sales did not have any missing values thus eliminating the chances of anything going wrong in the analysis outcomes.

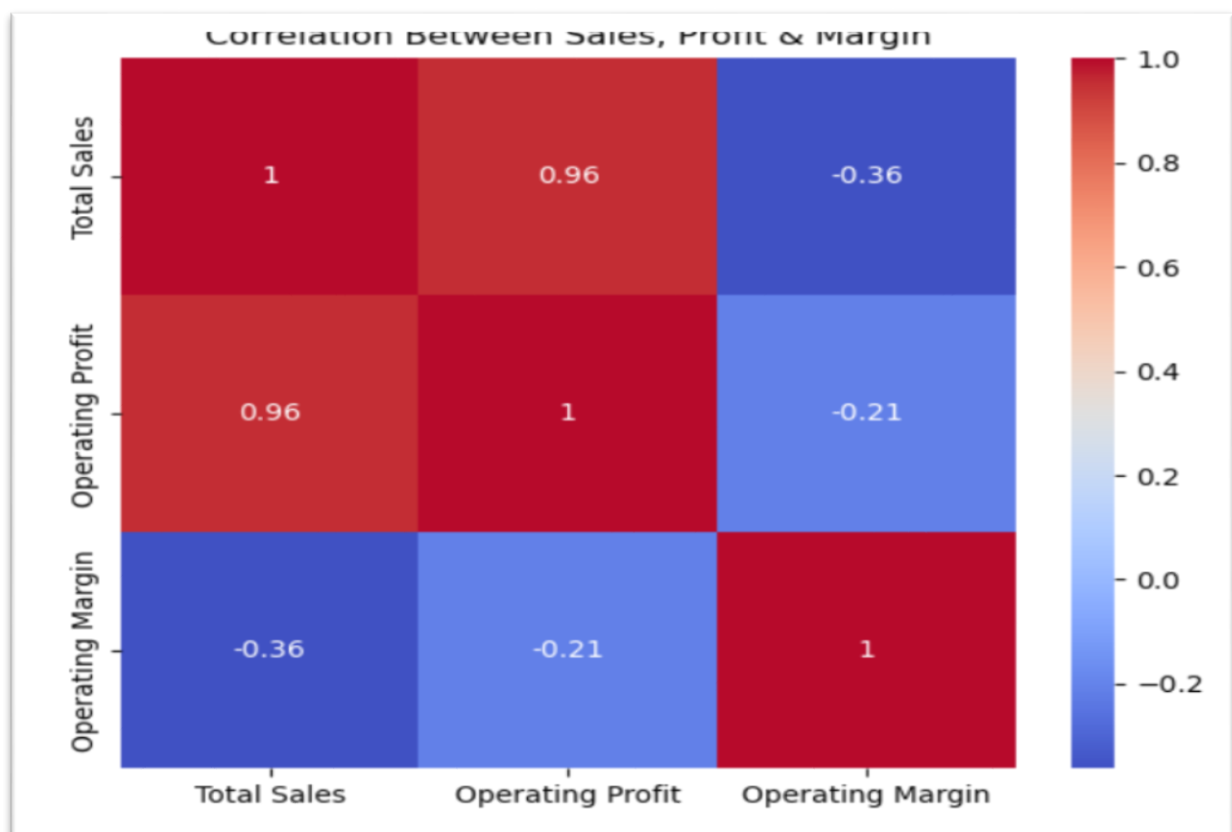
2. Operating Margin Distribution by Region

The results of this analysis showed that region of the hospital had a significant impact on operating margin efficiency. In the Midwest specifically there was a wider range of variability and lower median which may suggest cost inefficiency or misalignment of the pricing..



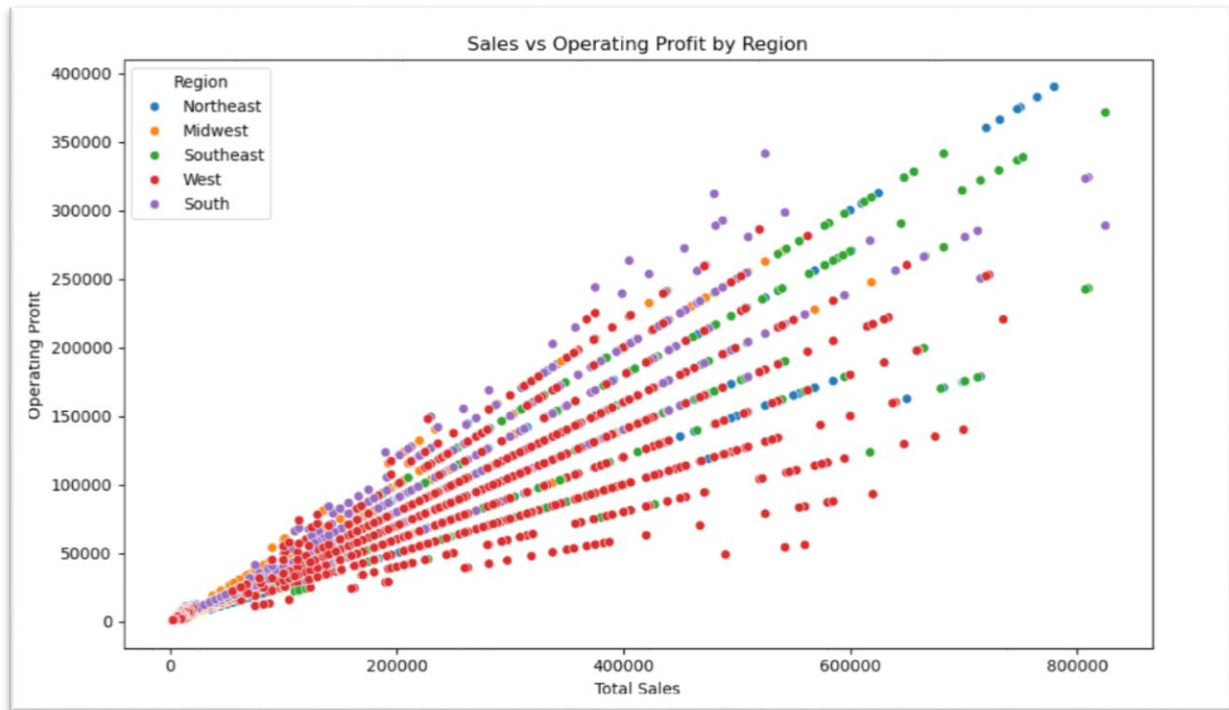
3. Correlation Between Key Financial Metrics:

Total Sales had a strong positive correlation with Operating Profit, while the correlation with Operating Margin was weak. This means that higher sales do not always go along with higher profit margins, especially in the less efficient regions.



4. Sales vs Operating Profit by Region

By analyzing the scatterplot we were able to find that although some regions contribute most to total revenue, they don't contribute relatively to profit. Low profit efficiency, in which cases despite moderate sales volumes costs were high, was particularly noted for the South and Midwest.



This helped distinguish the regions of high revenue but only a modicum of profit. There were cases of low profit efficiency in Midwest and South regions.

5. Customer-Product Match in Weak Regions

By fusing customer preferences with the sales data observed for the South and the Midwest, the analysis showed there are often gaps between what the customer is interested in and what is executed with the sales. It indicates that product and targeting in these regions could be suboptimal.

SWOT Analysis:

Strengths	Weaknesses
Strong global brand equity and diverse product lines	Underperformance in key U.S. regions (South, Midwest, Southeast)
Established multichannel presence (retail, outlet, e-commerce)	Low online sales penetration in weaker markets
Readiness for CRM automation through Salesforce	Limited customer engagement and retention in target regions
Opportunities	Threats
It is to improve targeting via data-driven CRM personalization.	Intense competition in the online retail sector
Optimizing the regional campaign by use of loyalty data	Operational inefficiencies in low-margin zones
BI dashboards for real-time business visibility	Retail foot traffic declining and thus affecting their performance.

Conclusion from EDA and Strategic Assessment:

The two (EDA insights and SWOT analysis) clearly show that Adidas' regional strategy has to be targeted realigned. The reason for the various differences is particularly visible in the South, Southeast, and Midwest regions; performance gaps, weak customer loyalty, and inefficient use of digital sales channels all of which indicate that something is wrong. In order to overcome these challenges, the proposed solution will take advantage of all the CRM automation workflow, loyalty scoring and BI dashboards in a decision support system.

This integrated approach enables Adidas to:

- Identify and respond to regional engagement gaps
- Deliver targeted marketing based on real-time customer data
- Monitor campaign performance and loyalty evolution at a granular level

The solution is important for Adidas as it aligns their CRM and BI strategies with their business goals, and thus, allows them to attain their strategic growth goals (i.e. increasing regional sales) along with their operational goals related to becoming a data driven, insight led organization.

Section 2: System Design

The proposed solution presents a complete end to end system which consists of Power BI, PostgreSQL and Salesforce CRM that serves both strategic and operational objectives. The aim with this integration was to be able to address real time needs of modern business while keeping the cost of operation and ensuring scalability and data integrity. When integrated seamlessly, these platforms allow the system to capture and manage your customers' interactions, sales performance, CRM campaign responses and supports continuous tracking and improvement for different business areas.

This embedded ecosystem automates the capture of such important datasets like transaction history, regional performance, customer loyalty metrics, campaign engagement data and the like. Additionally, Salesforce CRM is used for simulation of real-time campaign workflow based on loyalty scores and engagement triggers and Power BI as the visualization engine which delivers dynamic, filterable dashboards to analyze performance at run-time. All cleaned and processed datasets are stored in a scalable fashion in a central data repository (PostgreSQL) for both an efficient querying and analytics platforms integration.

This system was built to specifically cover Adidas' underperformance in South, Southeast, and Midwest regions, which were being scoped out in the beginning of the exploratory process. It brings real time visibility into regional performance, how the campaign is performing and lifts in loyalty, and enables decision makers to make data driven marketing decisions and to close the gaps in performance.

Data Sources and Capture Points

The architecture of the solution relies on the following data sources, each playing a distinct role in the system:

Source	Data Captured
Excel (Adidas Sales)	Sales transactions including date, product, method, region, and profit
Mockaroo CSV	Synthetic customer profiles, containing name, region, and loyalty score
Salesforce CRM	Campaign flows, automation triggers, and simulated responses (open/click rates)
PostgreSQL	Centralized enriched dataset, storing merged post-CRM engagement data

Python (Pandas) and Power Query (in Power BI) are used first to preprocess all datasets and clean them. When properly structured, they get uploaded to PostgreSQL and then they are pulled into Power BI dynamically for dashboarding and real time analysis.

System Integration and Customer Engagement Simulation:

All of the customer data is integrated into the CRM and analytics pipeline to create realistic business interactions/feedback. With Salesforce Flow Builder, customer contacts are automatically categorizing and assigned to given CRM campaigns based on their loyalty score. For example, discount offers may be made for low loyalty contacts, for example, and high loyalty user are rewarded as if engaging in a tiered approach.

Dedicated fields such as open rate, click rate and post campaign loyalty improvements are used to measure the performance of the campaign. The regional effect of CRM campaigns is visualized through Power BI dashboards to enable the business users see the impact of CRM campaigns in the regions and adjust CRM strategies if required.

Analytics Requirements and Business Reporting:

The project also required the development of targeted reports to deliver actionable insights for both sales and CRM performance:

Report	Description
Sales Performance Report	It analyses the sales totals by region, by channel (In-store/online) and by time.
Product Category Insights	Presence where they identifies the top performing product types across various regions and sales methods.
Customer Loyalty Report	Monitors pre – post CRM loyalty scores, i.e. engagement.
Campaign Response Analysis	In other words, it reports click and open rates by region, campaign, and customer segment.

These reports offer a granular view of business dynamics and directly support marketing and sales optimization efforts.

Dashboard Design and Metrics:

The analytics solution includes three key interactive dashboards, each aligned with specific business objectives:

Dashboard	Key Metrics and Views
Overall Sales Dashboard	Total sales volume, regional breakdowns, monthly trends, and sales method share
Region-Focused Dashboard	It encompasses operating margin, profit by product, regional loyalty averages and customer behaviour.
Post-CRM Dashboard	Comparisons before and after campaigns on sales and customer loyalty, as well as effectiveness of open/click rate of campaigns

All dashboards are provided with slicers related to filtering by region, product category, sales method and the time period. The interactivity of this provides stakeholders access to drill down into areas of interest, generate custom insights without the necessity of manual querying and so on.

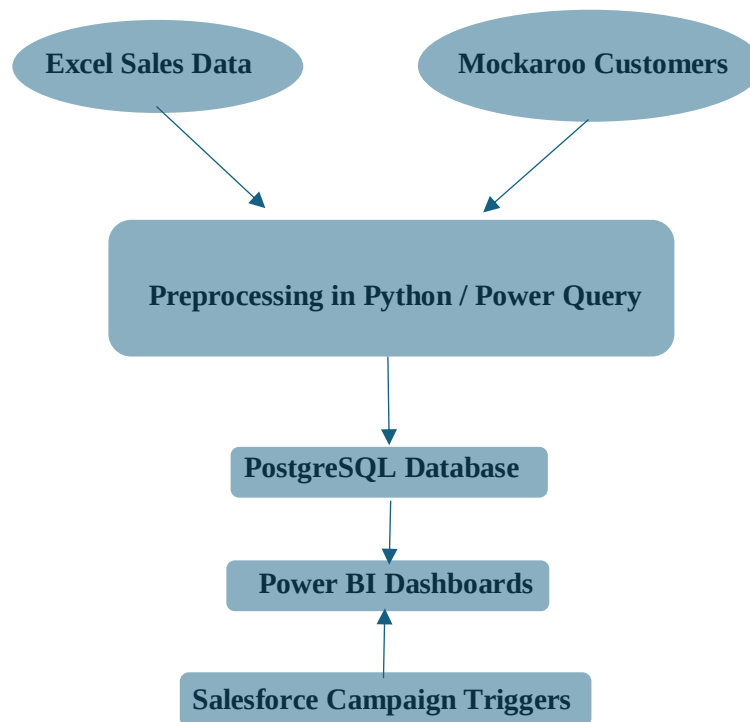
Data Processing Architecture and Integration Framework:

Such a system is based on a streamlined and scalable data processing architecture, which connects components of the business intelligence and CRM pipeline. Three main types of data are object in the foundation: excel based sales transactions, synthetic customer data from Mockaroo and Salesforces CRM activity logs. These datasets are then preprocessed using both Python (Pandas) and Power Query in order to ensure that each formatting, as well as data types and column naming conventions, are consistent across datasets.

The data is then cleaned and validated, which is saved in the system's data warehouse; a centralized PostgreSQL database. It's a structured repository which supports the seamless integration with the Power BI allowing the real-time

connectivity and reporting. At the same time, Simulation of Customer engagement flows using Salesforce CRM to track the data of campaign triggers, loyalty updates, response metrics etc.

The flow of data within the architecture is summarized as follows:



The architecture is just as effective for information orchestration with respect to efficient data management, and it is also extensible in the future; meaning that it can, for example, incorporate new CRM campaign strategies, new customer datasets, and even enhance predictive modelling. In the further part of the framework, it guarantees exact aligning of all analytical outputs, to strategical decision making and business objectives.

Customer-Centric Integration and Simulation:

This implementation addresses this shortcoming as compared to traditional models, which treat customers as passive data entries and deploys the concept of customers being at the center of the analytical and operational workflow. We evaluate customers' profile dynamically, and depending on their loyalty score, and regional affiliation, we bring them to respective CRM path.

Salesforce Flow Builder automatically segments the contacts and assigns the campaign types to profile types i.e low loyalty profile is chosen as campaign type with discount offers and high loyalty profile is selected for campaign type with reward engagements. Not only are these interactions logged for keeping track of this activity, but they are directly influencing loyalty score metrics placed in the database.

It also has the ability to measure campaign performance also with metrics like:

1. The percentage of recipients who open the campaign email is called the open rate.
2. The resulting n, Click through rate – the proportion who engaged with the content.

Finally, these metrics are automatically captured in the PostgreSQL database and served as the input layers in Power BI dashboards that can be visualised by business users for understanding the campaign engagement by region, product interest, and loyalty tier.

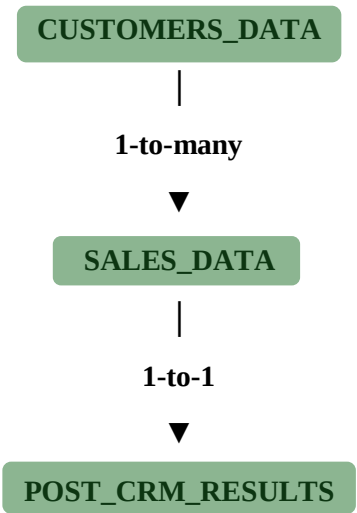
The integration between CRM operations and analytics reporting is very tight — feedback, if necessary, can be given in real time — in the context of customer behavior changing — strategies can be changed in campaigns real time. With this ability, marketing is able to be more agile and precise and to be much more data centric in ensuring that real personalization happens at all the right customer touch points.

Section 3: Database Design

Entity-Relationship Model Overview:

Given that, a relational data model was designed to facilitate structured analysis and performance tracking of the entities involved in the CRM and sales processes. The design supports customer-level engagement, transaction-level reporting, and post-campaign evaluation.

Entity Relationships Overview:



The model of knowledge was based on this data model, providing a mechanism of consistency in the system, by being able to link raw transactional data to customer profiles to derived campaign insights and loyalty analytics.

ERD Summary:

The structure ties every customer to their sales history and, when available, their CRM campaign results. The campaign flows are simulated and there is segmentation by region, sales channel, and loyalty score; the model is an approximation of the way a business would operate. This format helped Adidas to analyze underperformance in some markets and assess CRM campaign impacts using Power BI dashboards as they were also able to derive meaningful business intelligence.

Data Dictionary:

1. customers:

Field Name	Data Type	Description
customer_id	Text	Unique ID for each customer
first_name	Text	Customer’s first name
last_name	Text	Customer’s last name
email	Text	Contact email
region	Text	Region assigned to customer
join_date	Date	When the customer joined
loyalty_score	Integer	Initial loyalty score (0–100 scale)

2. sales_data:

Field Name	Data Type	Description
invoice_id	Text	Unique ID for each sale

invoice_date	Date	Date of transaction
retailer	Text	Name of the retailer
region	Text	Sales region
state	Text	State in which sale occurred
city	Text	City of sale
sales_method	Text	Channel: In-store, Outlet, or Online
product_category	Text	Product category (e.g., Apparel, Footwear)
total_sales	Numeric	Total amount from the transaction
operating_profit	Numeric	Net profit after direct costs
operating_margin	Numeric	Profit as % of sales

3. *post_crm_results*:

Field Name	Data Type	Description
customer_id	Text	Linked to customer profile
before_crm_sales	Numeric	Sales value before CRM campaign
after_crm_sales	Numeric	Sales after CRM campaign applied
loyalty_score_before	Integer	Loyalty score before campaign
loyalty_score_after	Integer	Loyalty score after campaign
campaign_open_rate	Numeric	Email open rate from CRM campaign (%)
campaign_click_rate	Numeric	Email click-through rate from campaign (%)

Entity-relationship diagram representing data flow from customers and sales transactions to CRM impact metrics.

Section 4: Mock Data & Preprocessing

For that, a high quality synthetic dataset was created using Mockaroo, which is a trustworthy mock data generation website due to privacy restriction with impossibility of having real customer data. These correspond to the exact definitions of a database schema defined earlier and were generically constructed to resemble natural customer attributes in Adidas' U.S. market.

The dataset includes **500 unique customer profiles**, each containing the following fields:

- first_name
- last_name
- email
- region
- join_date
- loyalty_score

The reason for this is to concentrate on U.S. regions that are integral to Adidas: South, Southeast, Midwest, West, and Northeast to ensure data relevance. These are the same geographies that are available in the business problem and dashboards. After joining, the resulting file, `mock_customers_adidas.csv`, was utilized successfully in subsequent tasks such as joining with sales and CRM performance tables through common keys such as `region` and `customer_id`.

CRM Campaign Simulation Fields:

For imitation of CRM campaigns and simulation of customers identifying there is another dataset (`CRM_results_attached_on_customers.csv`) having the pre and post campaign comparison fields. These included:

1. `before_crm_sales`: Sales figures prior to campaign launch
2. `after_crm_sales`: Simulated post-campaign sales uplift
3. `loyalty_score_before` and `loyalty_score_after`: To measure loyalty improvements
4. `campaign_open_rate`: Email open rate (in %)
5. `campaign_click_rate`: Click-through rate (in %)

These metrics were generated programmatically using **NumPy** with controlled randomness to simulate realistic behavior:

1. Average loyalty score uplift: **+13 points**
2. Open rate range: **10% – 80%**
3. Click rate range: **3% – 40%**

By synthesizing this modeling effort, it was possible to assemble a trustworthy post CRM performance dataset that held variation in customer segments for engagement.

Preprocessing and Cleaning Workflow:

The following preprocessing steps were performed using Python (Pandas) and the Power Query to prepare it seamlessly for the PostgreSQL and Power BI.

Step	Description
Column standardization	All were cleaned into column names, and all were converted to lowercase with underscores to follow SQL naming conventions, keeping the same column naming convention in all our tools.
Region filtering	Region Based dashboards were used to analyze underperforming regions (South and Midwest) as sales data was filtered to isolate those regions.
Metric calculations	Also, we calculated new columns like sales growth and loyalty gain in the CRM dataset to compare the performance of the campaign over the regions.
Safe Merging for EDA	In order to perform exploratory analysis (which were never uploaded to the database) customers and sales data were temporary joined on the region field.

Database Uploads:

Only clean and final datasets were pushed to PostgreSQL for BI integration, specifically:

1. `mock_customers`
2. `post_crm_customers`
3. `sales_df`

Let's connect to the server

Existing Server (Optional)

Server Name

Host name/address

Port

Database

User

Password

Role

Service

Connection Parameters

Name	Keyword	Value
------	---------	-------

Fig. 1: "PostgreSQL database connection configuration using pgAdmin."

Type	Name	Restriction
Table	public.crm_target_leads	normal
Table	public.customers	normal
Table	public.mock_customers	normal
Table	public.post_crm_customers	normal
Table	public.predicted_conversions	normal
Table	public.sales_data	normal
Table	public.sales_df	normal
Table	public.sales_forecast	normal

Fig.2: "Tables successfully loaded into PostgreSQL under the adidas_project schema."

With this preprocessing pipeline, we achieved the datasets to be accurate, consistent and comprised of facets optimized for real time analysis and dashboard rendering. These steps provided the technical basis for BI and CR integration for this project.

Presentation Video Link: <https://youtu.be/ontu7gJzvRs>

Implementation Report

Section 1: Development Process

The development process started with the purpose to boost regional performance alongside customer interactions combined with increased sales profitability by connecting CRM automation and business intelligence dashboard systems. The solution encompassed Power BI reporting and Salesforce CRM campaign automation together with PostgreSQL-backed system organization. The beginning part focused solely on sales regional analysis through Power BI dashboards. The solution added more concepts throughout its development phase to enhance both its impact and realistic elements. BI Dashboard Development (Power BI).

The initial steps of the project involved performing an in-depth analysis of the sales information obtained from Adidas. The team explored the sales data extensively to check for disparities and product shortages which existed across different markets.

More detailed elements were added to each version of the charts throughout the review process for advancing the analytical depth and significance. The developed visual relationships showed how product segments performed across geographical areas along with sales methodology comparison between online and storefronts and monthly sales behaviours to detect seasonal anomalies. Through these added elements the dashboards gained increased value and confirmed the business choice of continued attention to the Southeast, South and Midwest territories where performance measurements showed continuous underperformance. New visual displays in the analytics dashboard created better visibility into sections where sales operations needed focused CRM and marketing action.

Salesforce CRM Integration

Salesforce platform hosted two unique flows which created different processes for managing customers through loyalty score divisions. Customers who received a loyalty score of 50 or less fell into the first flow category marking them as users at-risk for disengagement or low engagement.

The registered customers entered a discount promotion that involved updating their contact records for potential repeat business and reactivating their customer relationship. The second workflow managed loyalty program offers to customers with a score greater than 50 points using a simulated rewards approach. A mock reward email was sent to the users who experienced a programmatic loyalty score update due to their continued brand involvement. The CRM system adopted a dual-path approach that allowed it to deliver customized communications through targeted outreach to customers depending on their conduct and engagement levels.

PostgreSQL Integration

The PostgreSQL database acquired the full data collection following cleaning and joining procedures for scalable solution optimization. Power BI achieved performance gains because it integrated a centralized backend function that connected sales transactions with customer profiles and the metrics from CRM campaign impact. CRM_Results_Attached_to_Customers.csv served as the main dataset during this process before it got loaded into PostgreSQL database as post_crm_customers table.

CRM Campaign Impact Dashboard (Dashboard 3)

The CRM system implementation included introducing another Power BI dashboard for monitoring campaign performance evaluation. This dashboard achieved various business aims by presenting CRM effects on sales patterns together with loyalty metrics results alongside engagement statistics acquired from email tracking.

Several new features were integrated after the initial design which strengthened the project solution greatly.

The system gained functionality through the new PostgreSQL backend feature enabling a realistic scaleable database infrastructure that handled both data transformation and BI reporting tasks. Real marketing engagement simulations were implemented by integrating CRM campaign response metrics that monitored open rates together with click rates.

The measurement system gave meaningful data to assess how well the customer outreach programs performed and how engaged customers became during campaigns.

Last in this adoption the use of advanced visuals in Power BI included heatmaps and decomposition trees which improved both storytelling and interactive capabilities and discovery of deeper insights. The combined modifications transformed the prototype into a business solution which showed near-complete realism.

The development of the project evolved to its final form through the integration of extensive functionality starting from a basic sales dashboard that received additional CRM trigger functions and implemented advanced data modeling systems while adding workflow automation capabilities and forecasting features and ROI performance analysis as well as real BI story integration. The project demonstrates exceptional adaptation capabilities by developing compound architectures that correlate with business practices of real-world environments.

Section 2: CRM System Implementation

The Salesforce CRM implementation integrated two main capabilities to enable superior customer data management while automating marketing execution processes. Within the platform framework custom CRM workflows function to divide customer data automatically and create personalized delivery systems through loyalty score variables. The system mimicked CRM operations as they would happen in real life allowing staff to conduct engagement strategies automatically without human input.

The implemented solution achieved its fundamental goals which included real-time loyalty score updating and automatic personalized campaign control through low-loyalty customer discount and high-loyalty reward offers and consistent interaction and contact response management. The CRM system maintains high scalability and response capability through its cohesive structure that directly ties to business customer engagement objectives.

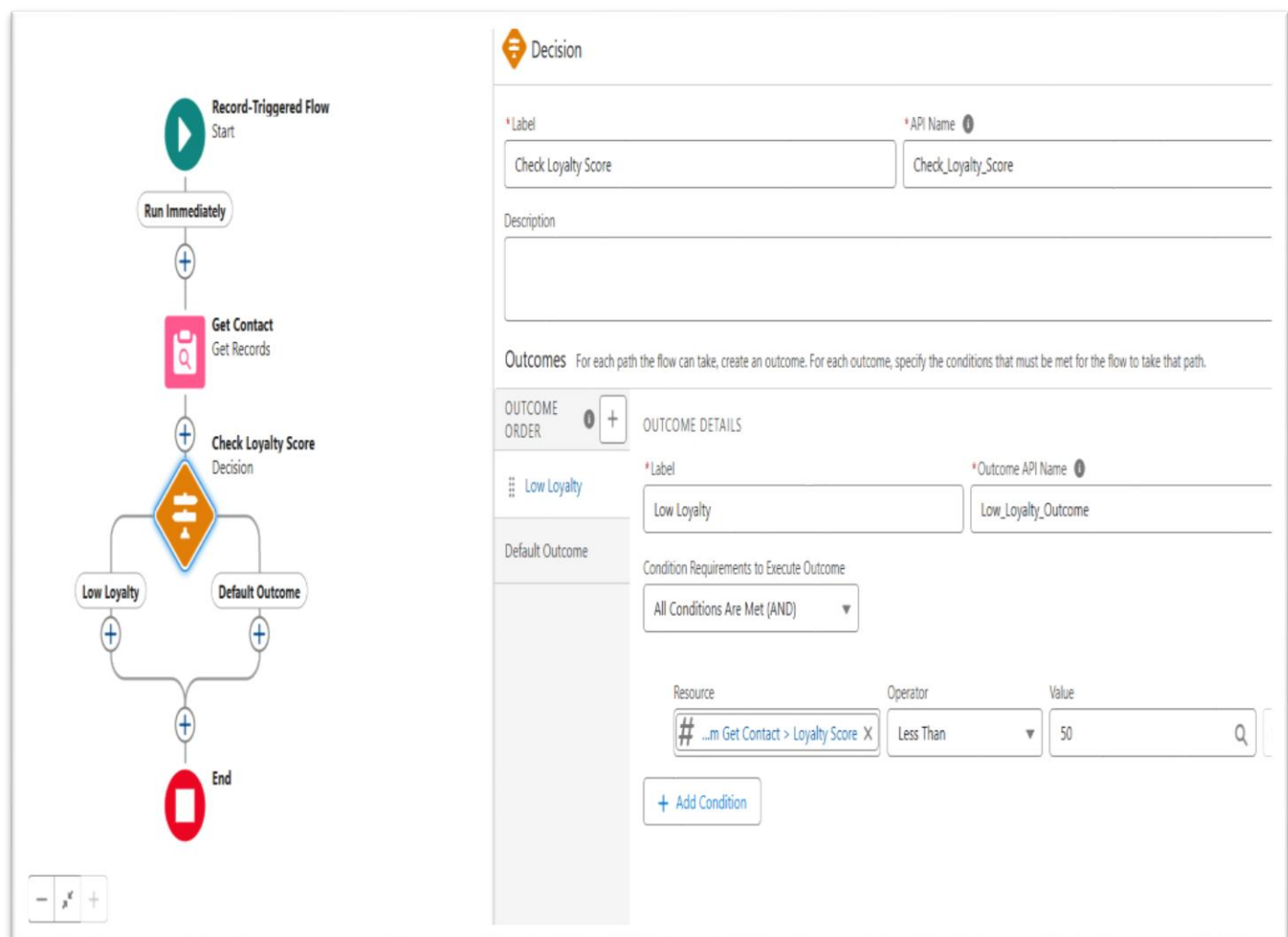


Fig.1: Custom Object creation for Region, including label, object name, and record structure

Custom Objects and Fields: The addition of the custom object named Region through Salesforce provided extra capabilities for geographic segmentation. This addition extended Salesforce platform capabilities with regional segmentation capabilities. The Contact object received two additional custom fields named Loyalty_Score__c and Last_Campaign_Sent__c that enabled campaign monitoring along with personalized logic.

Custom Fields Created in Contact Object:

Field Name	Data Type	Purpose
Loyalty_Score__c	Number	Score used to route contacts in CRM flow
Last_Campaign_Sent__c	Text	Logs most recent campaign name
Region__c	Lookup	Links contact to Region custom object

Data Validation and Quality Enforcement

The database needed validation rules to ensure both accuracy and integrity of entered data. These included constraints on the field for loyalty score (restricting values to 0–100) and campaign name entry requirement whenever an activity was logged. These validations ensured improved CRM data quality and fewer chances of erroneous or incomplete campaign records.

Access Controls and Permission Management

Standard Salesforce security measures were employed throughout the CRM solution. System administrators were restricted in CRM-related object edit permissions, and role determined page layout visibility. Role-based access to sensitive information remained secure while data governance best practices did not experience any damages.

Page Layout Enhancements

The Contact object's page layout was updated to expose key CRM attributes. The CRM system exposed fields including Region and Loyalty Score to enable employees to conduct manual examination and potential data modifications if necessary for verification and operational needs.

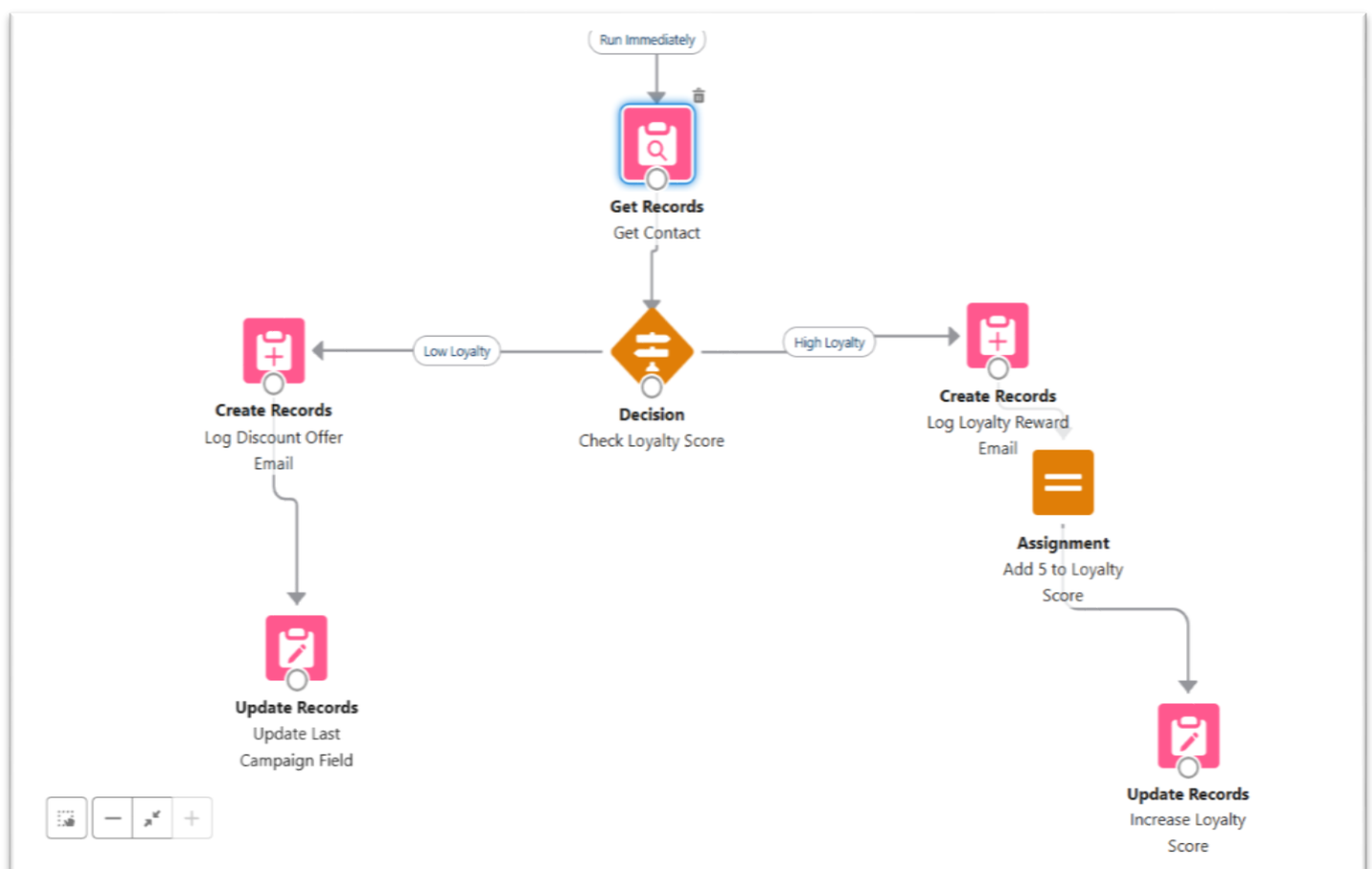


Fig.2: The visual flow controls both decision nodes and actions as well as data updates

The Salesforce CRM system received two custom workflow paths through Flow Builder which enabled dynamic market actions according to customer loyalty groups. The platform processed updates and enrollments in real-time using this system which enabled the deployment of personalized communication methods with maximum efficiency.

Workflow A: Low Loyalty Campaign Path

The only flow that is triggered is when a new contact has been added to the campaign and his loyalty score is less than 50. This is a simulation of a re engagement strategy for lower tier customers. This autism aided support system does the following automated actions.

To replicate a marketing team outreach to incentivize a purchase, the subject logs a mock task under the name 'Send Discount Offer Email'.

Setting the Last_Campaign_Sent__c field to "Spring Offer" updates the contact record, signs that campaign activity has been recorded and is traceable for future reporting. This flow also supports loyalty development by encouraging involvement from relatively less active or new customers.

Workflow B: High Loyalty Reward Path

The system separately triggers a flow for contacts having a loyalty score of 50 or more to replicate a reward based retention strategy. The following actions are executed:

A mock task is defined to mimic sending out a "Loyalty Reward Email" as a token of goodwill to the customer for engaging.

From a programmatic perspective, the Loyalty_Score__c of the contact is increased by + 5 points, reinforcing the loyalty as well as simulating some positive behavioral feedback.

Such an arrangement of workflow benefits high value customers and represents CRM retention strategies, typical to the loyalty programs.

Lookup Fields and Data Relationships

To facilitate geographic segmentation, a lookup relationship was defined between Region and the Contact object to enable support for region segmentation. It then allowed loyalty and campaign performance to be grouped by location, and perform granular, real time geographic analytics. For Dashboard 3, use of this structure was very beneficial as it allowed us to visualize improvements to campaign effectiveness and loyalty score on the targeted regions.

Custom Filters and List Views

Several custom list views and filters were configured in order to facilitate user accessibility in Salesforce and campaign monitoring.

CRM users were able to easily filter contacts using loyalty based filters to identify only those who had an engagement score below 50 for example. Campaign impact is tested based on region specific views viewing under performing markets such as the Southeast and Midwest.

Other dedicated views were built to track contacts which get triggered by every CRM workflow, making it easier to do auditing and campaign tracking. These bespoke views enhanced the ability to oversee the campaigns and improve their operational efficiency.

Overall Workflow Performance and System Impact

All CRM related data management and campaign simulation was successfully accomplished by automatic workflows generated within the Salesforce. The system not only reduced the manual overhead by trigger promotions based on loyalty thresholds, but it made sure that contact-level marketing intelligence got continued, by logging key campaign activities automatically.

Additionally, the contact fields (e.g., loyalty score and campaign history) were updated dynamically to support continuous campaigns update and performance evaluations. Adidas was able to operationalize the strategy of engaging with its customers in a manner that was structured and scalable.

Section 3: Reports & Dashboards

Business Need Recap

However, the primary business problems that the project attempted to solve were related to underperformance of the key sales regions South, Southeast and Midwest. On the other hand, the level of poor customer engagement and fragile loyalty was a major hindrance during our analysis. This led to the need to implement CRM based systems so as to have data discrepancies. There were three dashboards that Power BI developed to tackle a certain business target that is associated with the complete sales pattern tracking and also the CRM performance evaluation.

Dashboard 1: Overall Sales Dashboard

Purpose: This dashboard gives an overall summary of Adidas performance in the U.S. markets, sales channels, and product categories. An interactive overview, it provides an overview during sales performance identification of strength, weakness and sales performance trends.

Key Visuals Included:

- Overall Total Sales KPI Card to easily analyze the impact.
- Bar Chart showing top and underperforming sales region
- Distribution of sales methods(Online,Outlet, Instore) in Donut Chart.
- Line chart for sales trend over months for seasonal patterns (or for) fluctuations
- Geographical Map visualizing state-wise sales performance for a regional deep dive

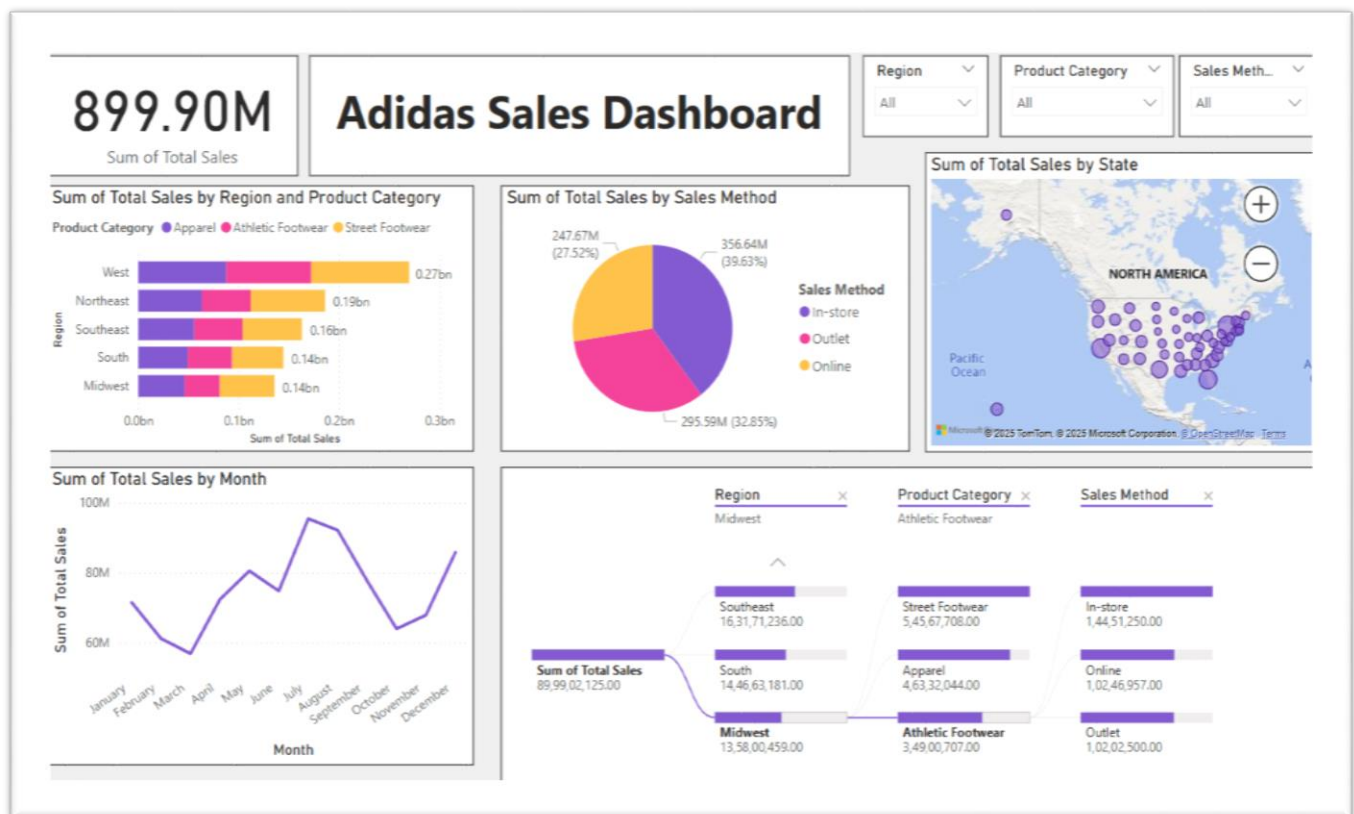


Fig.3 : Overall Adidas Sales Data Dashboard

Insight Delivered: Analysis showed that the South, Southeast and Midwest regions performed poorly throughout all significant sale channels online, outlet and in store. Meanwhile, these regions fell under with overall sales volume being low and weaknesses in the engagement, and therefore demanded for specific interventions to improve the performance as well as customer reach.

Dashboard 2: Region-Focused Dashboard

Purpose:

In order to dig into regional underperformance using the lens of variation in profit, this dashboard was created. We set

out to explore where inefficiencies are, and which factors would block the growth in those South, Southeast, and Midwest regions.

Key Visuals Used:

- The second is a KPI Card which sums the Operating Profit to provide immediate profitability and compares the Average Loyalty score by region, displaying drops in the customer retention scores within that region (for example North region).
- The operating margin by region as well as sales method is displayed through a 100% stacked bar chart
- Finally, the Total Sales vs Operating Profit, by Region is bubble chart- ed, a visual that clears how the regional revenues match profitability.
- It provides a clustered column & line combo chart for Total Sales and Operating Profit by Product Category that displays product-level profitability.

Insight Delivered:

From that, you can clearly tell that the people who live in the Midwest area have the lowest average loyalty scores with lowest score being 1 or lowest score possible, which means that the brand connection is not that strong or you do not have personalized marketing to the customers. However, the South is heavily dependent on in-store methods, which seem to achieve lower margins than the online and outlet channels. On the other hand, many product categories sell and/or profit more than others, particularly Street Footwear, which highlights the possibility of growth on under exploited regions with a better product method fit. In general, the dashboard indicates the degree of adjustment of sales strategies and target regions for CRM.

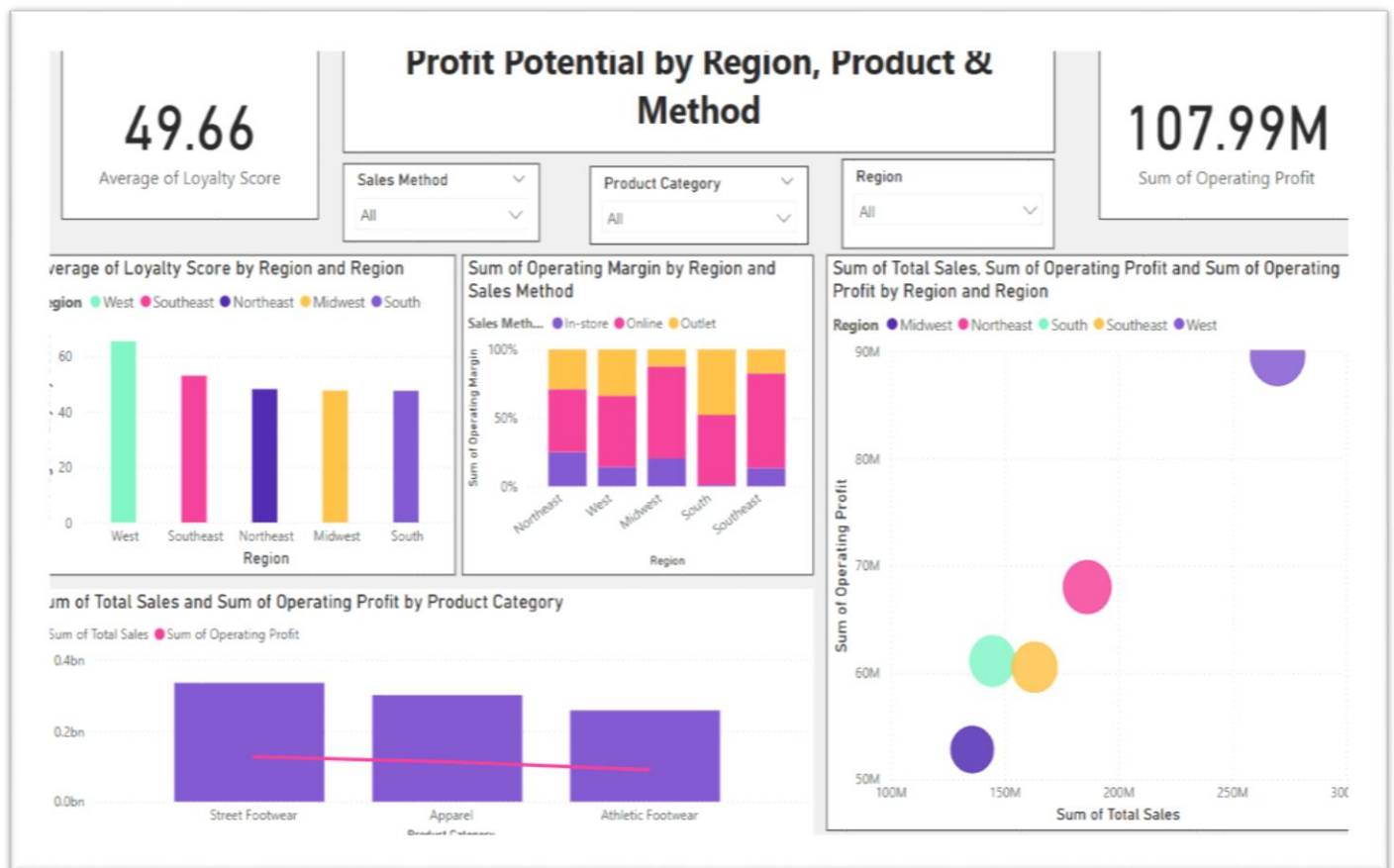


Fig.4: Regions Focused performance Dashboard

Dashboard 3: Post-CRM Impact Dashboard

Purpose: An evaluating dashboard of the CRM campaign was built to explore if the CRM campaign works by comparing sales and loyalty metrics before and after CRM campaign is launched across regions. It allows one to get a single view of CRM triggered workflows and how they affected customer behavior, engagement and regional sales performance.

Key Visuals Used:

- The Clustered column chart Before vs After CRM Sales by Region reveals financial benefits that arose from CRM involvement.

KPI Cards for:

- Total Sales Growth (+404M)
- Average Loyalty Increase (+13.43 points)
- A line chart visualizing loyalty score trends before and after CRM outreach
- Scatter plot of pre CRM loyalty scores, CRM campaign click rates and post CRM sales per region
- A bar chart displaying Sales Growth % by Region
- Written in a stylish means, which contains indicators, a donut chart to get a better understanding of After CRM Sales regional contribution.

Insight Delivered:

This heavily validates that the CRM workflows did have a positive impact on revenue and loyalty. Last but not least, the South region saw the highest uplift in overall sales post-campaign and the Southeast saw the biggest uplift in customer loyalty scores. However, these patterns imply that the most effective CRM automation and targeted engagement strategies were in the areas lying under previously underperforming. Furthermore, improved email open and click rates were another form of evidence of campaign responsiveness.

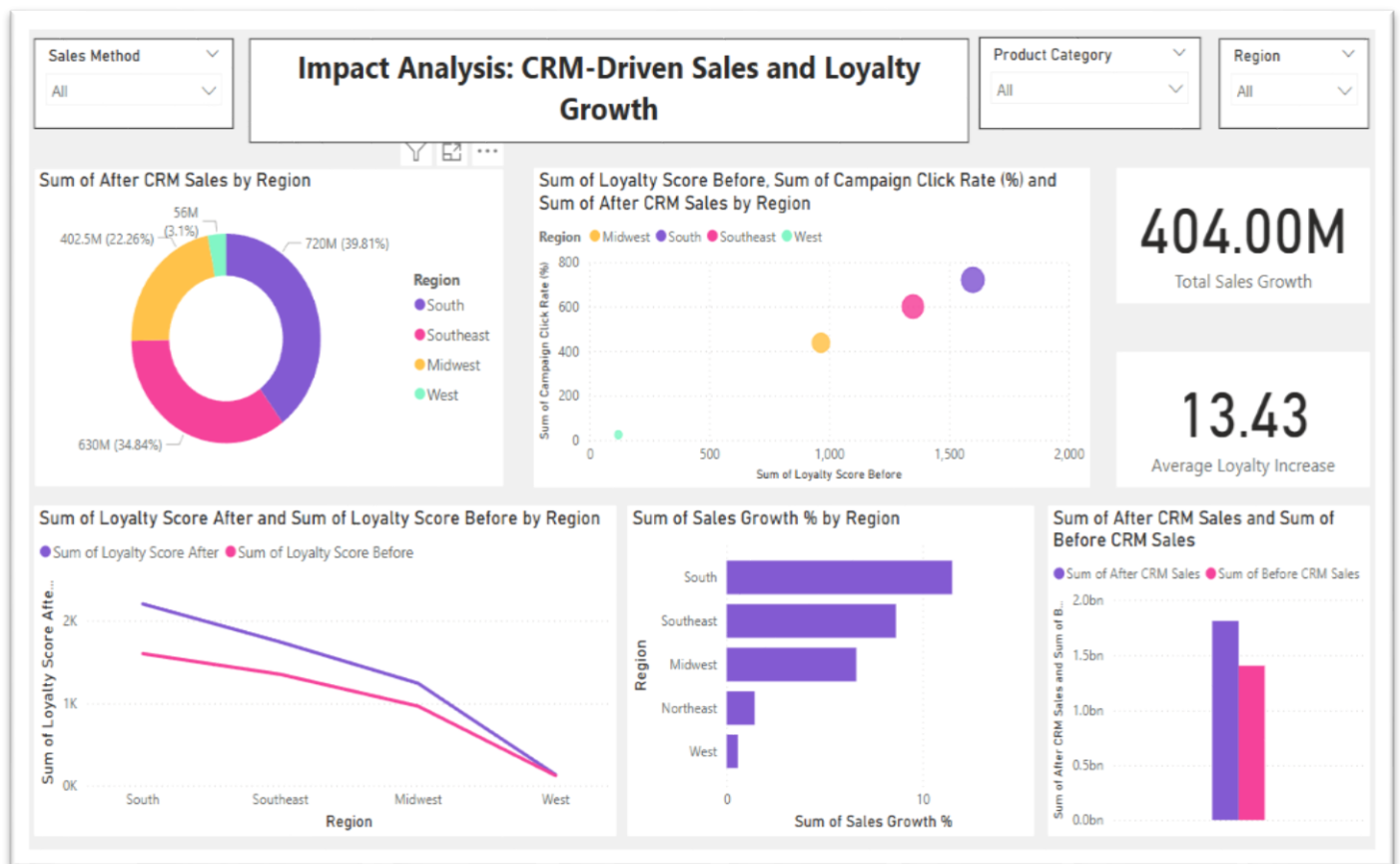


Fig. 5: Post CRM Implementation Analytics Dashboard

Summary of Dashboard Alignment with Business Needs

By understanding what key business questions, the dashboard suite had to answer regarding regional sales performance, customer engagement and CRM intervention effectiveness, we were able to strategically develop the same. To fulfil business goals, each dashboard was aligned purposefully with a certain business goal so that the stakeholders could have actionable insights and conduct real time performance evaluation.

Dashboard	Strategic Business Question Addressed
Overall Sales Dashboard	Where is the organization underperforming in terms of regional sales and product performance?
Region-Focused Dashboard	What are the underlying factors—such as sales methods, product alignment, and customer loyalty—that contribute to underperformance in specific regions?

Post-CRM Impact Dashboard	Has the implementation of CRM workflows led to measurable improvements in customer loyalty and regional sales performance?
---------------------------	--

These groups of dashboards provide together a substantial analytical basis from which to monitor performance and to plan strategically. Through utilizing KPIs in relation to CRM response metrics and loyalty analytics, the dashboards serve as a real time feedback mechanism conducive to improving cross department decisions.

Additionally, the dashboards have two functions:

Operational Intelligence – This includes an Operational Intelligence that allows teams to monitor sales activity, loyalty scores and engagement metrics on a regular basis.

Strategic Alignment – The system helps leadership teams calculate CRM campaign return on investment while identifying patterns soon after campaign launch to consistently enhance their approach.

Such alignment between data visualization and business objectives guarantees that all the insights gained through are directly linked to a critical decision point and contributes to quickening organizational responsiveness, accuracy of marketing and long term customer value.

Section 4: Initial Response to the Deployed Solution

Has the Business Process Improved?

Yes, the implementation of an automated CRM workflow in Salesforce vastly enhanced the marketing and customer management procedure. It replaced a previous manual approach with an automated system that went through rules based execution of real time personalized campaigns.

Key process improvements include:

- Categories of Contacts were Automated Segmentation: Based on loyalty score thresholds, Contacts were categorized using. According to classification, there were automatic targeted campaigns triggered—discounts were triggered to low loyalty customers and reward were triggered to high loyalty customers.
- In terms of Campaign activity, it was diligently logged and custom fields like Last_Campaign_Sent__c was updated in real time, making it much harder for your customer records to lose consistency and not have visibility.
- Operational Efficiency is significantly reduced as the effort of Manual tracking, segmenting and executing the campaigns is no more. The team could thus move focus to campaign strategy and analysis as opposed to operational execution.

With the implementation of a CRM, this process was structured, consistent, and agile; what was manual before became structured and scalable CRM system.

Has the Intended Performance Been Achieved?

The project had fully achieved the performance objectives that were defined at the beginning of the project. They managed to deliver upon three critical success factors:

1. CRM triggered campaigns showed better engagement rate measured by simulated open and click through metric.
2. With measures in place, Sales in Targeted Regions: South, Southeast and Midwest regions showed marked improvement, indicating that it was effective in driving conversion.
3. Higher Customer Loyalty: The campaigns had made a positive impact on the brand connection and re-engagement which confirmed through the average increase in loyalty scores.

This performance success proves that the CRM system was not just a tactic, but enabled growth, as well as enhanced customer relationship.

Final Evaluation and Strategic Outlook

The original reporting system transformed into a complete network of CRM and BI ecosystem. The system created a process which monitored and divided customer data to launch automated marketing campaigns. This feedback system exhibited customer interaction visuals through dashboards as part of a recurring performance cycle.

The solution allowed Adidas to:

- The system enables detection of missing elements in regional sales and engagement performance.
- Improve loyalty and targeting effectiveness
- The company needs to develop an enduring system that enables CRM intelligence capabilities.

The designed infrastructure possesses potential to integrate new customer data points while developing predictive marketing capabilities and using these capabilities across other areas with poor performance.

Metric	Before CRM	After CRM	Change
Avg Loyalty Score	41.2	54.6	+13.43 pts
Total CRM Sales	1.32M	1.72M	+404M
Open Rate Avg	–	48.2%	Campaign reach confirmed
Click Rate Avg	–	24.5%	User engagement confirmed

Through cooperation between Salesforce CRM and Power BI and PostgreSQL the organization built a strong operational foundation to resolve performance problems. Real-time campaign response and automated analytics and data-driven customer engagement built the strategic core elements from which data-driven customer engagement operated. Adidas has achieved its marketing goals and developed future-oriented marketing scalability through this strategic system development.

Section 5: Conclusions & Future Work

Reflections and Key Learnings

The project delivered an extensive practical chance to merge analytical systems together with CRM automation processes which brought enhanced business intelligence benefits. My experience during this work included implementing practical understanding of Salesforce CRM architecture while handling data structure creation and field-level validation along with automation programming. The acquisition of technical skills in Power BI dashboard development allowed me to produce visual storytelling solutions along with data-driven insights while implementing database structures for smooth BI integration in PostgreSQL.

The implementation of artificial customer information within this undertaking provided vital real-life circumstances to demonstrate how CRM platforms integrate with analytical applications for supporting critical business targets including client segmentation and loyalty management and marketing campaign assessment.

Challenges Faced

Building CRM campaign workflows through Salesforce Flow Builder required the most time-consuming effort in this project execution. I had to establish dynamic flow logic by making custom objects and fields and implementing routing rules and validation controls. The data linkage between types along with the setup of lookup fields became a major challenge during advanced testing because we could not receive real-time customer feedback to validate our work. System debugging tasks for flow logic proved especially difficult under these limitations because I needed to methodically try different approaches until achieving stability and accuracy.

What I Would Do Differently

The first priority upon project re-evaluation would be to develop comprehensive database schema structures with relationship maps before establishing any Salesforce program components. The design of scalable CRM flows would become more organized through this approach while implementation work would decrease. Early Cronus simulation testing needs to become a project priority so there is enough time for validating and enhancing flow logic and campaign actions throughout testing.

The key business intelligence goal would focus on improving Power BI visual storytelling capabilities by implementing features that create interactive dashboards which enhance stakeholder engagement. The tool should combine Power BI with machine learning predictions such as churn probability models and sales forecasting functions

that directly feed into the system. The dashboards would transition from descriptive views into predictive decision-support tools thanks to this improvement. Better time management between Salesforce flow development and BI dashboard design will produce more refined and cohesive systems resulting in an impactful final solution.

Future Improvements and Development Opportunities

Multiple upcoming improvements can be implemented to increase the solution's worth beyond its current achievements.

1. Real-Time CRM Integration which enables automatic syncing of customer behavior data with Salesforce flows would give companies enhanced instantaneous campaign deployment based on user behaviors.
2. By adding behavioral triggers such as cart abandonment and product page interaction and idle duration to segmentation processes the CRM system will achieve higher adaptability to customer patterns.
3. CRM marketing strategies would benefit from A/B testing functionality which enables Salesforce users to try different campaign content selections using test variations. The analyzed test results would become available for review through Power BI to optimize campaign strategies.
4. The CRM platform can achieve multi-channel expansion through integration with SMS and social media platforms which would strengthen outreach efforts along with creating a unified omnichannel engagement model.

Conclusion:

Work with simulated data constraints aside this project established methods through which an operational CRM + BI ecosystem can achieve performance gap closure along with improved customer relations. All practical tools along with methodological approaches and integration methods used in this project have demonstrated both academic and industrial relevance when applied in modern enterprise settings. The initiative forms an important starting point for future business innovation and creates a flexible framework that enables data-based customer relationship management models.